

Week 5, Lecture 2 WOD – Answer Key

Writing the Likelihood

NRES 746 – Instructor Answer Key (not for student distribution)

This WOD took the place of the Wednesday lecture, so students met the idea of a likelihood for the first time here, working alone and then in groups. It is ungraded. Questions with a single correct answer are answered directly below. The open-ended items (1d, 2d, 2e, the group question, 4c, 5c, and the muddiest-point block) get discussion points instead of an exact-match rubric.

Everything below is computed in the chunks, so the numbers can be checked against a student's paper. The handout's four nests are the same throughout.

```
y <- c(1, 1, 0, 1)          # outcomes for nests 1-4
cover <- c(30, 20, 10, 30)  # percent sagebrush cover
```

What students should walk away with

1. A likelihood is the probability of the data you actually observed, viewed as a function of the parameter with the data held fixed: $\mathcal{L}(p | y) = P(y | p)$. The formula is the same one as in a probability calculation. What changes is which quantity is treated as known.
2. Under independence, the likelihood of several observations is the **product** of the individual probabilities. The **log** turns that product into a sum, which is what you compute in practice.
3. A likelihood is **not** a probability distribution over the parameter. The values don't sum to 1 (2d), and even the area under the whole curve is not 1.
4. When the probability depends on a covariate, the likelihood is a function of the *parameters of that relationship* (a and b). The data (the outcomes and the covariate values) are what is held fixed.
5. The R version of all of this is `-sum(dbinom(y, size = 1, prob = p, log = TRUE))`.

Part 1: One observation

1a. For a single nest with success probability p :

$$P(y = 1 | p) = p \quad P(y = 0 | p) = 1 - p$$

1b. The standard answer is

$$P(y | p) = p^y(1 - p)^{1-y}$$

When $y = 1$ the second factor is $(1 - p)^0 = 1$ and the answer is p . When $y = 0$ the first factor is $p^0 = 1$ and the answer is $1 - p$. This is the Lab 3 binomial pmf with one trial, because $\binom{1}{y} = 1$. An equally correct form with no exponents is $yp + (1 - y)(1 - p)$, which also gives p and $1 - p$.

```
bern <- function(y, p) p^y * (1 - p)^(1 - y)
c(bern(1, 0.75), bern(0, 0.75))      # 0.75 and 0.25
```

```
## [1] 0.75 0.25
```

Common mistakes to watch for: writing a piecewise or “if” rule (not allowed by the prompt); writing $p^y(1-p)$ and forgetting the exponent on the second factor; writing $y \cdot p$ alone (this is 0 when $y = 0$, which is not a probability of the outcome).

Definition (given in the handout). No answer needed.

1c. For nest 1, $y_1 = 1$, so $\mathcal{L}(p \mid y_1 = 1) = p$:

$$\mathcal{L}(0.75 \mid y_1 = 1) = 0.75 \quad \mathcal{L}(0.25 \mid y_1 = 1) = 0.25$$

$p = 0.75$ makes the observed outcome three times as probable as $p = 0.25$ does. **Common mistake:** giving 0.75 for $\mathcal{L}(0.25 \mid y_1 = 1)$ or mixing up which of y and p is plugged into which slot.

1d. Discussion points. In $p^y(1-p)^{1-y}$, y is the data and p is the parameter. In a Lab 3 probability calculation, the parameter p was known (given) and the outcome y was the thing we asked about. In a likelihood, the outcome is known (already observed) and p is the unknown that we vary. The formula is the same. What changes is which symbol is fixed and which one we let vary. This is the single most important idea in the WOD, so listen for whether groups state it in their own words.

Part 2: Several observations

2a. With independent nests sharing one success probability p , the joint probability factors into a product:

$$\mathcal{L}(p \mid y_1, y_2, y_3, y_4) = \prod_{i=1}^4 p^{y_i} (1-p)^{1-y_i}$$

Common mistakes to watch for: adding the individual probabilities instead of multiplying (this mixes up “or” and “and”); leaving out the product symbol and writing the four terms with no connection between them.

2b. With $y = (1, 1, 0, 1)$ there are three successes and one failure:

$$\mathcal{L}(p \mid \text{data}) = p \cdot p \cdot (1-p) \cdot p = p^3(1-p)^1, \quad j = 3, k = 1.$$

The order of the nests doesn’t matter, only the counts of successes and failures. **A variant answer to accept:** a student who treats the data as “3 successes out of 4 nests” and writes $\binom{4}{3}p^3(1-p)$ has used the binomial pmf with $N = 4$. The factor $\binom{4}{3} = 4$ is a constant that doesn’t depend on p , so it multiplies every value in the table by 4 (12/256, 64/256, 108/256) but does not move the peak. This is worth raising in the group discussion if a group hits it.

2c.

```
lik <- function(p) p^3 * (1 - p)
p_vals <- c(0.25, 0.5, 0.75)
data.frame(p = p_vals, likelihood = lik(p_vals), in_256ths = lik(p_vals) * 256)
```

```
##      p likelihood in_256ths
## 1 0.25 0.01171875         3
## 2 0.50 0.06250000        16
## 3 0.75 0.10546875        27
```

$$\mathcal{L}(0.25) = \frac{1}{64} \cdot \frac{3}{4} = \frac{3}{256} \approx 0.012 \quad \mathcal{L}(0.50) = \frac{1}{8} \cdot \frac{1}{2} = \frac{16}{256} = \frac{1}{16} = 0.0625 \quad \mathcal{L}(0.75) = \frac{27}{64} \cdot \frac{1}{4} = \frac{27}{256} \approx 0.105$$

2d. The three values sum to $46/256 \approx 0.18$, not 1.

```
sum(lik(p_vals)) # 0.1797
```

```
## [1] 0.1796875
```

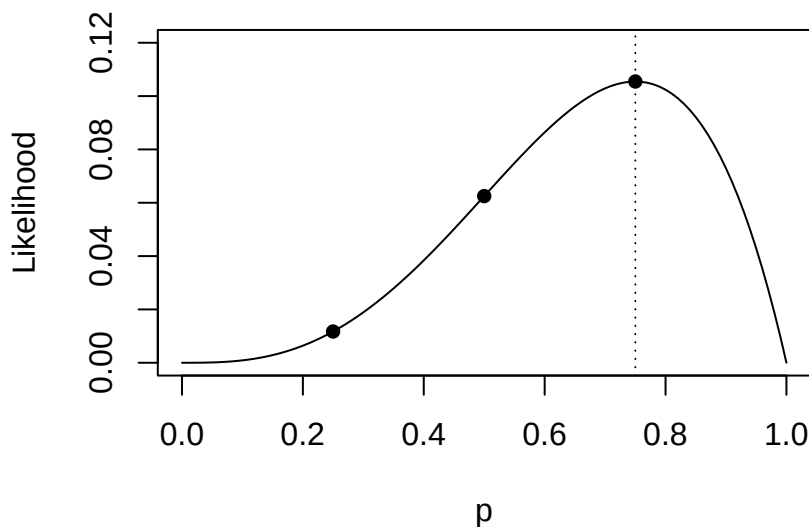
```
integrate(lik, 0, 1)$value # area under the whole curve = 1/20
```

```
## [1] 0.05
```

Even the area under the whole curve from 0 to 1 is only $1/20 = 0.05$. *Discussion points.* The probabilities in a Lab 3 calculation sum to 1 across the *possible outcomes* for a fixed p . A likelihood is evaluated across *possible parameter values* for fixed data, and nothing forces those to add up to 1. A likelihood value is therefore **not** “the probability that p is 0.75”. It is the probability of *these data* if p were 0.75, and it is only meaningful relative to other values of p . Students often assume the three numbers should sum to 1 because they came from probabilities: the sum-to-1 question is designed to surface that.

2e. The curve rises to a maximum at $p = 3/4$ and falls to 0 at both $p = 0$ and $p = 1$:

```
par(mar = c(4, 4.5, 1, 1))
curve(x^3 * (1 - x), 0, 1, xlab = "p", ylab = "Likelihood", ylim = c(0, 0.12))
points(p_vals, lik(p_vals), pch = 16)
abline(v = 0.75, lty = 3)
```



The peak is at $p = 3/4$, which is also the sample proportion of successes (3 of 4). Setting the derivative to zero confirms it: $\frac{d}{dp} p^3(1-p) = p^2(3-4p) = 0$ at $p = 3/4$. Most students will guess 0.75 from the raw data (“three out of four hatched”), which is exactly the point: the peak of the likelihood agrees with intuition here. Students are not expected to do the derivative. It previews the maximum likelihood work in Week 6.

Part 3: The log-likelihood

3a. Take logs, then use $\log(uv) = \log u + \log v$ and $\log(u^c) = c \log u$:

$$\ell(p) = \log \prod_{i=1}^n p^{y_i} (1-p)^{1-y_i} = \sum_{i=1}^n [y_i \log p + (1-y_i) \log(1-p)]$$

Grouping terms gives the equivalent form $\ell(p) = (\sum y_i) \log p + (n - \sum y_i) \log(1-p)$. For the four nests that is $3 \log p + \log(1-p)$, and

```
3 * log(0.75) + log(0.25)      # ell(0.75)
```

```
## [1] -2.249341
```

```
log(27 / 256)                  # log of the likelihood at 0.75: same number
```

```
## [1] -2.249341
```

Common mistakes to watch for: writing $\log(a + b)$ or trying to “distribute” the log over a sum; losing the $(1 - y_i)$ exponent and writing $y_i \log p + \log(1 - p)$.

3b. Each term is about 0.5, so the product is $0.5^{2000} = 2^{-2000}$. With $\log_{10} 2 \approx 0.30$:

$$\log_{10}(2^{-2000}) = -2000 \times 0.30 = -600, \quad \text{so the product is about } 10^{-600}.$$

(The exact value is 10^{-602} ; the rounded log gives 600.) That is far below 10^{-308} , so a computer cannot store it and the product **underflows to exactly 0**. The sum from 3a stays representable: each term is $\ln 0.5 = -\ln 2 \approx -0.69$, so the sum is $2000 \times (-0.69) \approx -1380$ (exactly -1386).

```
prod(rep(0.5, 2000))          # underflows to 0
```

```
## [1] 0
```

```
sum(log(rep(0.5, 2000)))      # -1386.29
```

```
## [1] -1386.294
```

```
c(exact_log10 = -2000 * log10(2), with_rounded_log = -2000 * 0.30)
```

```
##      exact_log10 with_rounded_log
##      -602.06      -600.00
```

The moral is that a likelihood over many observations is a product of many numbers below 1, so in practice you work with the log-likelihood, a sum of moderately sized negative numbers. *Optional connection:* the birthday-problem code in the idea bank made the same move to log space.

Group check-in

Group question. *Sample answers; accept any biologically sensible one.* Nests aren’t independent when they share something that affects hatching beyond the modeled probability:

- The same hen renests after a failure, or a single hen’s condition affects all her nests.
- Nests in the same area face the same predators, and a predator that finds one nest raises the odds of finding its neighbors.
- Nests in the same year or the same storm share weather.
- Nests at the same lek or on the same ridge share habitat quality that cover doesn’t fully capture.

The general idea: shared, unmeasured causes make the outcomes of neighboring nests correlated. This foreshadows hierarchical models later in the course.

Part 4: Adding the covariate

4a. For nest 2, cover is 20%, so $a + bx = -2.2 + 0.11(20) = -2.2 + 2.2 = 0$:

$$p_2 = \frac{1}{1 + e^{-0}} = \frac{1}{1 + 1} = 0.5$$

```
plogis(-2.2 + 0.11 * cover)    # exact predictions for all four nests
```

```
## [1] 0.7502601 0.5000000 0.2497399 0.7502601
```

The other three round to the handout's values: 0.7503 for nests 1 and 4, and 0.2497 for nest 3.

4b. Each nest contributes p_i if it hatched and $1 - p_i$ if it failed. Nests 1, 2 and 4 hatched and nest 3 failed:

$$\mathcal{L} = 0.75 \times 0.5 \times (1 - 0.25) \times 0.75 = 0.75 \times 0.5 \times 0.75 \times 0.75 = \frac{27}{128} \approx 0.211$$

```
0.75 * 0.5 * (1 - 0.25) * 0.75 # by hand with the rounded p's
```

```
## [1] 0.2109375
```

```
prod(dbinom(y, 1, plogis(-2.2 + 0.11 * cover))) # with the exact p's (0.2112)
```

```
## [1] 0.211157
```

Common mistake to watch for (this is what the prompt warns about): using $p_3 = 0.25$ for the failed nest instead of $1 - p_3 = 0.75$. That gives $0.75 \times 0.5 \times 0.25 \times 0.75 = 0.070$, which is a third of the right answer.

4c. The best constant- p value from 2c is $27/256 \approx 0.105$ at $p = 0.75$. The covariate model gives $27/128 \approx 0.211$. That is **twice** as likely, so the data support the covariate model more. *Discussion points on caution.* Any of these is a good reason:

- Only four nests, so a factor of 2 could easily arise by chance.
- The covariate model has two parameters and the constant model has one. More flexible models can fit data better even when the extra parameter is meaningless (this returns as model selection later).
- The pair $(a, b) = (-2.2, 0.11)$ was handed to us. We never asked whether some other pair does even better, or how much the answer would change under a different pair.
- A likelihood ratio of 2 says nothing yet about uncertainty.

Common mistake to watch for: concluding “cover matters” without qualification, or comparing 0.211 to a constant- p likelihood that isn't the best of the three (for example $p = 0.5$).

4d. The likelihood for n nests, with $p_i = 1/(1 + e^{-(a+bx_i)})$, is

$$\mathcal{L}(a, b \mid y_1, \dots, y_n, x_1, \dots, x_n) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}, \quad p_i = \frac{1}{1 + e^{-(a+bx_i)}}.$$

Written out in a and b only (using $1 - p_i = 1/(1 + e^{a+bx_i})$):

$$\mathcal{L}(a, b) = \prod_{i=1}^n \left(\frac{1}{1 + e^{-(a+bx_i)}} \right)^{y_i} \left(\frac{1}{1 + e^{a+bx_i}} \right)^{1-y_i}.$$

Underline (fixed data): y_i , x_i , and n . **Circle (what the likelihood is a function of):** a and b .

Common mistake to watch for: circling p_i . The p_i are not free quantities: each is determined by a , b and the nest's own x_i . The unknowns are the two parameters of the relationship. This is the step from Part 2 (one parameter p) to a real model.

Part 5: From math to R code, on paper

5a.

```
nll_const <- function(p, y) {
  -sum(dbinom(x = y, size = 1, prob = p, log = TRUE))
}
```

The blanks are $x = y$ (the observed outcomes), `size = 1` (one trial per nest, which makes the binomial a Bernoulli), and `prob = p`. `dbinom` is vectorized over x , so it returns one log-probability per nest, and `sum()` adds them, which is exactly the sum from 3a. The minus sign turns the log-likelihood into its negative. **Common mistakes to watch for:** $x = 1$ or $x = p$ (confusing which argument is the data); `size = length(y)` (this would treat all the nests as one binomial experiment, which changes the function); forgetting the minus sign.

5b. `exp(-nll_const(0.5, y))` undoes both the log and the minus sign, returning the likelihood of $p = 0.5$:

```
exp(-nll_const(0.5, y))      # 0.0625 = 1/16
```

```
## [1] 0.0625
```

```
nll_const(0.5, y)           # = log(16) = 2.77
```

```
## [1] 2.772589
```

The answer is $1/16 = 0.0625$, which matches the $p = 0.50$ row of the 2c table (16/256). **Common mistake to watch for:** stopping at `nll = log 16  $\approx$  2.77` without converting back with `exp(-...)`.

5c. The smallest negative log-likelihood is at $p = 0.75$, which agrees with the peak in the 2e sketch.

```
p_grid <- seq(0.05, 0.95, by = 0.01)
nll_vals <- sapply(p_grid, function(p) nll_const(p, y))
p_grid[which.min(nll_vals)]          # 0.75
```

```
## [1] 0.75
```

```
exp(-min(nll_vals))            # 0.1055 = 27/256
```

```
## [1] 0.1054688
```

The minimum of the negative log-likelihood sits at the same p as the maximum of the likelihood because log is increasing and the minus sign flips the curve. *Why negative?* Any of these is a good answer:

- Optimizers such as R's `optim()` **minimize** by default (as students saw with the SSR in Lab 2), so we hand them a quantity that is smaller when the fit is better.
- Since likelihoods for discrete data are at most 1, log-likelihoods are at most 0. Their negatives are nonnegative “badness of fit” numbers, like the SSR.
- Maximizing $\ell(p)$ and minimizing $-\ell(p)$ are the same problem.

5d (optional).

```
cover <- c(30, 20, 10, 30)
```

```
nll_cover <- function(par, y, cover) {
  p <- 1 / (1 + exp(-(par[1] + par[2] * cover)))
  -sum(dbinom(y, size = 1, prob = p, log = TRUE))
}
```

```
exp(-nll_cover(c(-2.2, 0.11), y, cover))    # 0.2112, matches 27/128 = 0.2109 (rounded p's)
```

```
## [1] 0.211157
```

The value is approximately 0.211, matching the hand answer to 4b. The small difference (0.2112 against 0.2109) is because 4b used predicted p values rounded to 0.75, 0.5 and 0.25.

What is still fuzzy?

Read these for where the class needs help. Likely muddiest points, and where to address them:

- **Likelihood versus probability** (the 2d sum-to-1 issue). This is the most likely one. Consider opening next Monday's maximum likelihood lecture by returning to it.
- **Why the product, why the log** (independence and underflow).
- **What “the data are fixed” means** when the data are still random variables in the model.
- **Whether the peak of the likelihood is “the answer”** and how sure we can be of it (uncertainty). This comes in with confidence intervals in Weeks 6-7.
- **The R syntax** in Part 5 (`dbinom` arguments, `sapply`), which is expected to be new for some students.